

Part 1 — One Problem, Three Lenses (Modules 1–4)

INFO 521 · Machine Learning Foundations · project overview

How to read this

This overview is the project brief. The milestone notebooks carry the authoritative Satisfactory criteria, each stated at the top of its own notebook. Grading mechanics, including the gate and bundle structure, are on the course Assessment page.

Course-schedule placement. Part 1 of the project: assigned Week 1, due Week 5, the midpoint. Milestones on the module arc: 1.1 Least Squares (M1 to M2, due Wk 3), then 1.2 Maximum Likelihood (M3, due Wk 4), then 1.3 Bayesian (M4, due Wk 5; the conjugate Gaussian posterior is the Part 1 endpoint), then 1.4 Synthesis (due Wk 5). Gating: Checkpoint quizzes 1.1 to 1.3, each due the week before the milestone it gates.

The first half of the course project, due Week 5, the midpoint.

One clinical dataset, one prediction problem (age $\rightarrow$ systolic blood pressure), examined through three accreting lenses and then synthesized.

Milestones

1. **1.1 Least Squares** (M1–M2) — design matrix, normal equations, K-fold CV, L2. Play artifact: a 3×3 bias–variance sweep over polynomial orders $\{1, 3, 9\}$ and $\lambda \in \{1e-3, 1e-1, 1e1\}$. Gated by Checkpoint quiz 1.1 (due Week 2; milestone due Week 3).
2. **1.2 Maximum Likelihood** (M3) — Gaussian-noise generative model, MLE for $\mathbf{w}$ (shown numerically equal to the 1.1 OLS fit), $\hat{\sigma}^2$, predictive error bars. Play artifact: where the model is least sure, and why. Gated by Checkpoint quiz 1.2 (due Week 3; milestone due Week 4).
3. **1.3 Bayesian** (M4) — Gaussian prior, conjugate posterior, posterior-predictive credible intervals. Play artifact: prior-strength and data-amount sweeps showing collapse toward the MLE. Gated by Checkpoint quiz 1.3 (due Week 4; milestone due Week 5).
4. **1.4 Synthesis** — one panel, three lenses; argued compare-and-contrast; reproducible end-to-end. Structured peer review.

Through-line

Part 1 stays **single-predictor polynomial** the whole way (model order is the only complexity knob, exactly the Module 1–2 lecture arc). Expanding to the full multivariate feature set is one of the things **Part 2 adds**.

Satisfactory

Each notebook states its own criteria at the top. The project is complete when 1.1–1.4 are all Satisfactory and Checkpoint quizzes 1.1 to 1.3 are passed. See the Assessment page.

Bridge to Part 2

1.3 leans entirely on conjugacy (closed-form Gaussian posterior). Part 2 opens by binarizing systolic BP into a **hypertension** label (ACC/AHA: $SBP \geq 130$ or $DBP \geq 80$) — conjugacy breaks, the posterior goes intractable, and approximate inference becomes necessary. The same binary label, predicted from the **non-BP** features (the blood-pressure columns are held out to avoid label leakage), then drives classification; dropping it entirely motivates clustering and PCA.